

VERIFICATION PLAN

(V-PLAN)

Project Title

Verification of AXI to AHB Bridge Using UVM

1. Introduction

The AXI to AHB Bridge is designed to enable communication between an AXI master and an AHB slave by converting AXI transactions into equivalent AHB transactions. Since AXI and AHB follow different protocol architectures and signaling mechanisms, it is necessary to verify that all transactions are translated correctly without loss of information.

The purpose of this verification plan is to define the verification objectives, test scenarios, and checking mechanisms used to validate the functionality of the AXI to AHB Bridge.

The verification environment has been developed using Universal Verification Methodology (UVM) and includes protocol-specific agents, reset agents, monitors, a virtual sequencer, and a scoreboard.

2. Verification Objectives

The following objectives were identified at the beginning of the project:

Objective 1 – Reset Verification

Verify that the bridge initializes correctly when reset is asserted and deasserted.

Expected Outcome:

- DUT enters a known state after reset.
 - Protocol signals are initialized correctly.
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Objective 2 – AXI Write Transaction Verification

Verify that AXI write transactions generated by the master are correctly converted into AHB write transactions.

Expected Outcome:

- AXI write address is translated correctly.
 - AXI write data is transferred correctly.
 - AHB write transaction is observed.
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Objective 3 – AXI Read Transaction Verification

Verify that AXI read transactions are translated into equivalent AHB read transactions.

Expected Outcome:

- AHB read operation is generated.
 - Read data is returned correctly to the AXI side.
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Objective 4 – Address Translation Verification

Verify that addresses observed on the AXI interface are correctly propagated to the AHB interface.

Expected Outcome:

- Address mismatch should not occur.
 - Address integrity must be maintained.
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Objective 5 – Data Integrity Verification

Verify that data transferred through the bridge remains unchanged during protocol conversion.

Expected Outcome:

- AXI write data matches AHB write data.
 - AHB read data matches AXI read data.
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Objective 6 – Response Verification

Verify correct handling of AHB responses.

Expected Outcome:

- OKAY responses are handled correctly.
 - ERROR responses are detected and propagated.
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Objective 7 – Wait-State Verification

Verify bridge operation when wait states are introduced by the AHB slave.

Expected Outcome:

- Transaction completion occurs correctly after wait-state insertion.
 - No data corruption occurs.
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3. Verification Methodology

The verification environment is developed using UVM.

The environment consists of:

- AXI Agent
- AHB Agent
- AXI Reset Agent
- AHB Reset Agent
- Virtual Sequencer
- Scoreboard

The AXI Agent generates master-side transactions while the AHB Agent models slave-side behavior. Monitors capture activity on both interfaces and send transactions to the scoreboard.

The scoreboard performs end-to-end checking by comparing transactions observed on the AXI side against transactions observed on the AHB side.

4. Verification Features and Test Scenarios

Feature	Verification Approach	Test Scenario
Reset Functionality	Reset Agent	reset_test
AXI Write Transaction	AXI Sequence	write_test
AXI Read Transaction	AXI Sequence	read_test
Address Translation	Scoreboard Comparison	write/read tests
Data Integrity	Scoreboard Comparison	write/read tests
AHB OKAY Response	Slave Response Sequence	response_test
AHB ERROR Response	Slave Response Sequence	error_test
Wait-State Handling	Slave Wait-State Sequence	wait_state_test
End-to-End Transaction Flow	Monitor + Scoreboard	Regression

5. Checking Strategy

Verification is performed using a combination of monitors, scoreboards, and waveform analysis.

Monitor-Based Checking

AXI and AHB monitors observe protocol activity and reconstruct complete transactions from signal-level activity.

Scoreboard-Based Checking

The scoreboard compares:

- Address information
- Data information

- Response information

Any mismatch between expected and actual behavior is reported.

Waveform Analysis

Waveforms are used to verify:

- AXI handshaking
 - AHB transaction generation
 - Response propagation
 - Reset behavior
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6. Coverage Goals

The following functionality should be exercised during verification:

Reset Coverage

- Reset assertion
- Reset deassertion

Transaction Coverage

- Write transactions
- Read transactions

Response Coverage

- OKAY responses
- ERROR responses

Wait-State Coverage

- Single wait state
- Multiple wait states

Protocol Conversion Coverage

- AXI Write → AHB Write
 - AXI Read → AHB Read
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7. Verification Completion Criteria

The verification effort will be considered complete when:

1. All planned test scenarios execute successfully.
2. The scoreboard reports no critical mismatches.
3. Reset functionality is verified.
4. Write and read transactions are verified.
5. Response handling is verified.
6. Wait-state scenarios are verified.
7. Waveforms confirm correct AXI-to-AHB protocol conversion.

8. Conclusion

This verification plan defines the approach used to validate the functionality of the AXI to AHB Bridge. The verification environment uses UVM-based stimulus generation, monitoring, and scoreboard checking to ensure that transactions generated on the AXI side are correctly translated and observed on the AHB side. The planned verification scenarios cover normal operation, reset behavior, wait-state handling, and error response conditions, providing confidence in the functionality of the bridge.